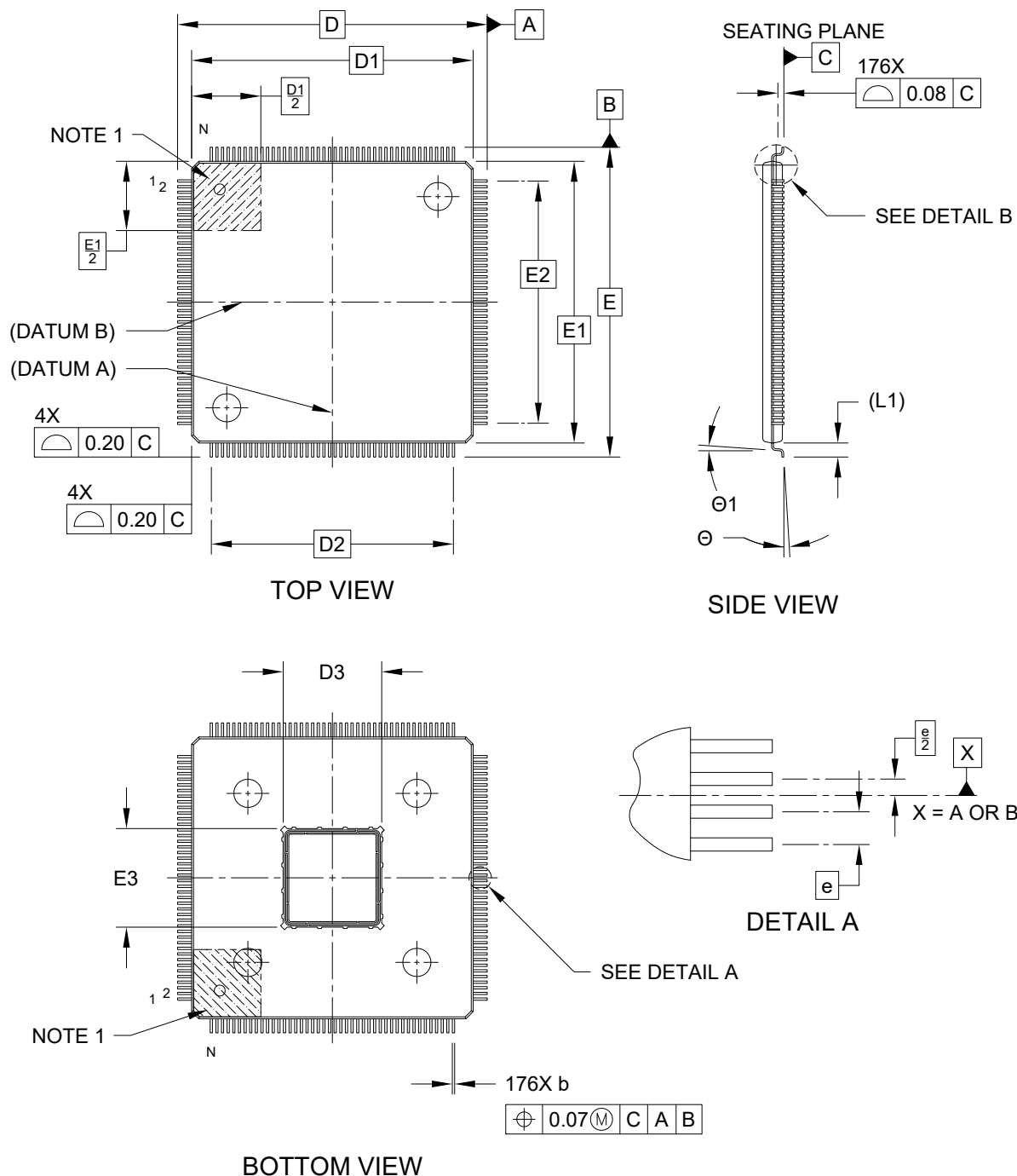


Packaging Diagrams and Parameters

176-Lead Low Profile Quad Flat Pack (2J) - 20x20x1.4 mm Body [LQFP] With 7x7 mm Exposed Pad

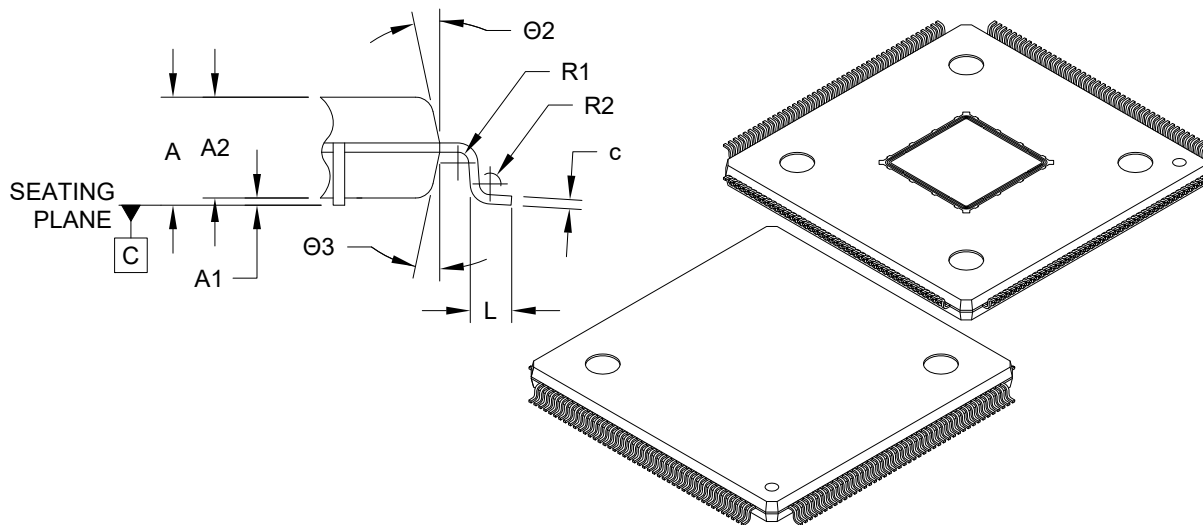
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

176-Lead Low Profile Quad Flat Pack (2J) - 20x20x1.4 mm Body [LQFP] With 7x7 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Leads	N	176		
Pitch	e	0.40 BSC		
Overall Height	A	-	-	1.60
Standoff	A1	0.05	-	0.15
Molded Package Height	A2	1.35	1.40	1.45
Overall Length	D	22.00 BSC		
Molded Package Length	D1	20.00 BSC		
Overall Lead Pitch	D2	17.20 BSC		
Exposed Pad Length	D3	6.90	7.00	7.10
Overall Width	E	22.00 BSC		
Molded Package Width	E1	20.00 BSC		
Overall Lead Pitch	E2	17.20 BSC		
Exposed Pad Width	E3	6.90	7.00	7.10

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Lead Width	b	0.13	0.16	0.23
Lead Thickness	c	0.09	-	0.20
Lead Length	L	0.45	0.60	0.75
Footprint	(L1)	1.00 REF		
Bend Radius	R1	0.08	-	-
Bend Radius	R2	0.08	-	0.20
Foot Angle	Θ	0°	3.5°	7°
Lead Angle	Θ1	0°	-	-
Mold Draft Angle	Θ2	11°	12°	13°
Mold Draft Angle	Θ3	11°	12°	13°

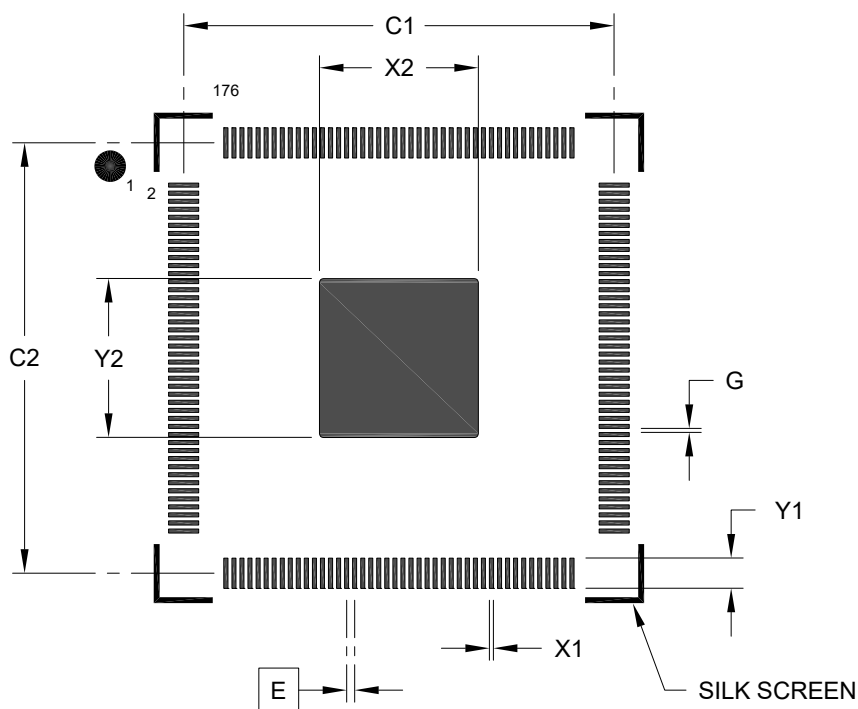
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensions D1 and E1 do not include mold protrusion. Allowable Protrusion is 0.25mm per side. D1 and E1 are maximum body size dimensions including mold mismatch.
- Dimension b does not include dambar protrusion. Allowable dam bar protrusion shall not cause the lead width to exceed the maximum b dimension by more than 0.08mm. Dambar cannot be located on the lower radius or the foot. Minimum space between protrusion and adjacent lead is 0.07mm for 0.40mm pitch packages.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

176-Lead Low Profile Quad Flat Pack (2J) - 20x20x1.4 mm Body [LQFP] With 7x7 mm Exposed Pad

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.40 BSC	
Contact Pad Spacing	C1		21.40	
Contact Pad Spacing	C2		21.40	
Contact Pad Width (X176)	X1			0.20
Contact Pad Length (X176)	Y1			1.50
Center Pad Width	X2			7.90
Center Pad Length	Y2			7.90
Contact Pad to Pad (X172)	G	0.20		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.